

IITISOC 2026 · DYNAMO TRACK 1 · PS 3

Hybrid Digital Twin for CO₂ Hydrogenation to Light Olefins

Final Evaluation — Physics-Informed Kinetics + Neural Surrogate + ASF Product Distribution + ASPEN Plus



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Python · TensorFlow / Keras · SciPy · NumPy · Pandas · Matplotlib

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Why CO₂ Hydrogenation to Light Olefins?



- Light olefins (ethylene, propylene, butylene) are core feedstocks for plastics, synthetic rubber and solvents.
- Catalytic CO₂ hydrogenation offers a route to valorize captured CO₂ into high-value chemicals, supporting carbon utilization goals.
- Industrially relevant design requires fast, accurate, physically-consistent process models — this motivates a hybrid digital twin.



\$200B+

Global annual market for light olefins

The Engineering Problem

Rigorous kinetic + reactor simulations are accurate but too slow for real-time optimization, flowsheet iteration, or process control.

Project Objectives

Build a hybrid (physics + ML) digital twin that is fast, accurate, and thermodynamically consistent



Physics-Based Kinetics

Model RWGS + Fischer-Tropsch reaction pathway with literature-validated rate laws.



Synthetic Dataset

Generate a kinetically-consistent dataset by solving reactor ODEs over wide T, P, feed-ratio space.



Neural Surrogate

Train a fast ML model that replaces the ODE solver for millisecond-speed inference.



ASF Consistency

Guarantee product-distribution physics using the Anderson–Schulz–Flory framework.



Aspen Plus Coupling

Embed the surrogate in a full flowsheet with recycle for plant-level digital twin.

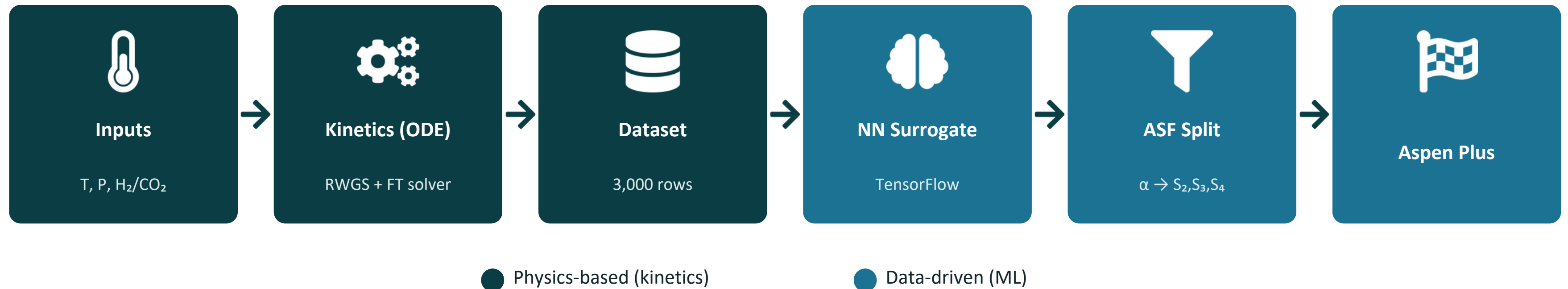


Design-Ready Maps

Deliver yield/selectivity parametric maps to guide reactor operating conditions.

Two-Stage Hybrid Architecture

Physics generates the training data; the network replaces the solver at inference time



Why hybrid? The ODE solver enforces chemical laws (mass balance, thermodynamics) when generating data, while the neural network learns the input–output map and delivers millisecond-speed predictions — fast enough for Aspen Plus parametric integration.

Reaction Mechanism & Kinetic Model

Brübach, Hodonj & Pfeifer (2022) — Fe/K/γ-Al₂O₃ catalyst, LHHW-type rate laws

STEP 1 — RWGS



$\Delta H = +41.2 \text{ kJ/mol}$ — endothermic

Converts CO₂ to CO; rate slows as the reaction quotient approaches Keq(T).



STEP 2 — FISCHER-TROPSCH



$\Delta H = -165 \text{ kJ/mol}$ — exothermic

CO undergoes chain growth on Fe sites, building the hydrocarbon product slate.

Fitted Kinetic Parameters (used directly in our ODE solver)

Parameter	Value	Description
EA,RWGS	115.0 kJ/mol	RWGS activation energy
EA,FT	67.8 kJ/mol	FT activation energy
aRWGS / aFT	16.3 / 9.07	H ₂ O / H ₂ inhibition terms
Tref	573.15 K	Arrhenius reference temperature

Reference

Parameter	Unit	Estimated value
$k_{\text{RWGS},300\text{ }^\circ\text{C}}$	$\text{mol h}^{-1} \text{g}^{-1} \text{bar}^{-1.5}$	8.13×10^{-2}
$k_{\text{FT},300\text{ }^\circ\text{C}}$	$\text{mol h}^{-1} \text{g}^{-1} \text{bar}^{-2}$	6.39×10^{-2}
$E_{\text{A,RWGS}}$	kJ mol^{-1}	1.15×10^2
$E_{\text{A,FT}}$	kJ mol^{-1}	6.78×10^1
a_{RWGS}	—	1.63×10^1
a_{FT}	—	9.07
b_{FT}	bar^{-1}	2.44

Anderson–Schulz–Flory (ASF) Model

Statistical chain-growth model governing Fischer-Tropsch product distribution

$$W_n = n(1 - \alpha)^2 \alpha^{n-1}$$

Reference

α range	Dominant product	Relevance
< 0.40	Methane (CH4)	Undesired
0.40 – 0.65	Light olefins C2–C4	OUR TARGET
0.65 – 0.80	Gasoline (C5–C11)	–
> 0.80	Waxes (C12+)	–

Dynamic α (Song et al. 2004, recalibrated)

We shifted the correlation's base term from 0.633 → 0.55 to target the light-olefin regime (α ≈ 0.40–0.65) instead of heavy hydrocarbons — consistent with Fe-K catalyst literature.

Synthetic Dataset Generation

Isothermal Plug-Flow Reactor ODEs, solved with `scipy.integrate.solve_ivp` (LSODA)

Plug-Flow Reactor Model

- Molar flow rates of CO₂, H₂, CO, H₂O and C1–C5+ species evolve with catalyst weight W.
- Solved with LSODA (rtol = 10⁻³, atol = 10⁻⁶) across wide ranges of T, P and H₂/CO₂ feed ratio.
- ASF weight fractions feed directly into the hydrocarbon-formation terms during integration.

Reference



Thermodynamic Consistency Check

$Q / K_{eq} \leq 1.05$ (5% numerical tolerance) for every simulated point.

100% of 3,000 rows passed

By construction — the RWGS rate naturally $\rightarrow 0$ as $Q \rightarrow K_{eq}$, preventing thermodynamic overshoot.

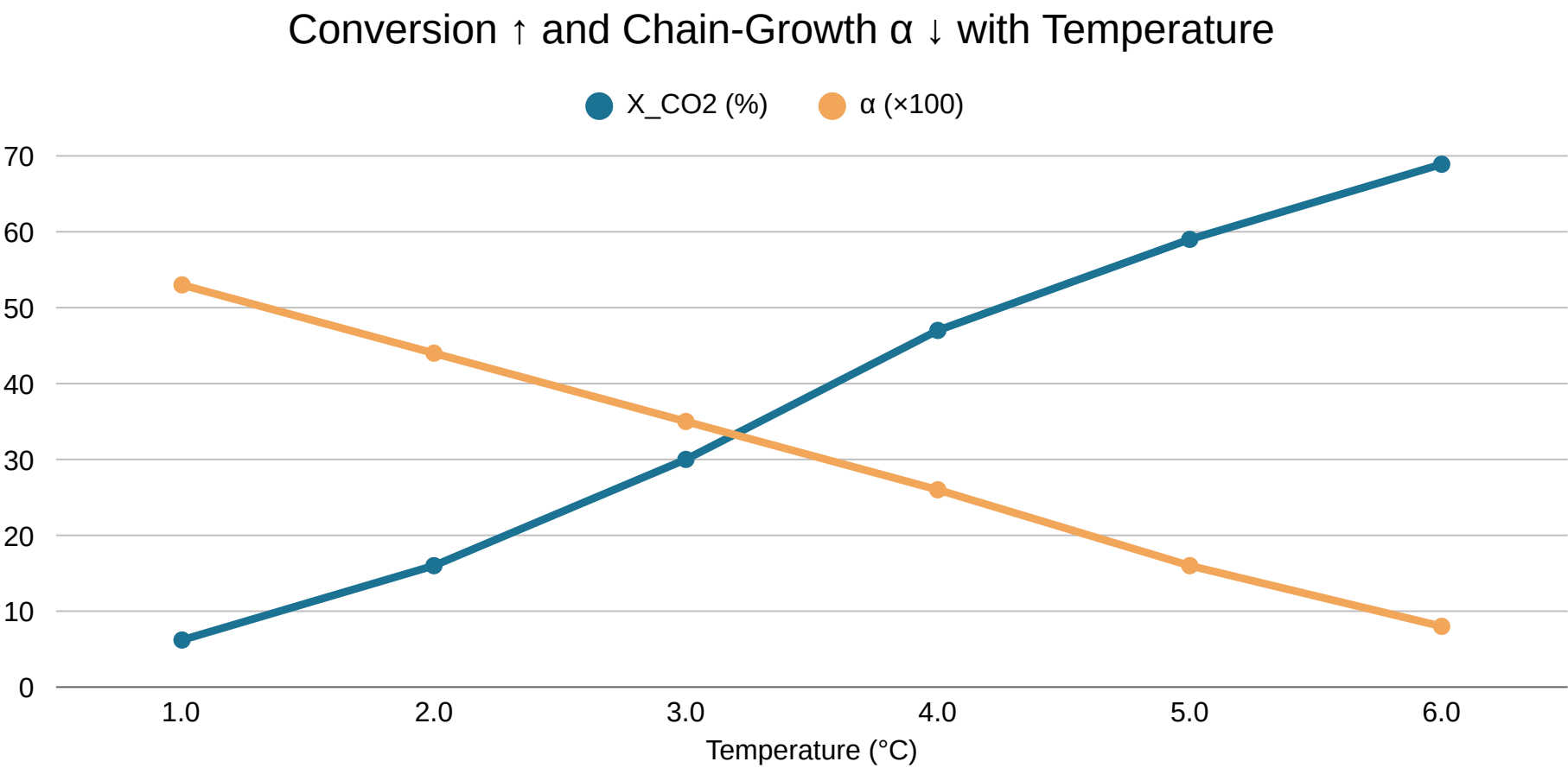
3,000 rows × 22 columns

- **Inputs:** T, P, H₂/CO₂ ratio, K_{eq}
- **Thermo Check:** Q/K_{eq}, thermo_ok flag, r_RWGS
- **Conversion:** X_CO2, α_outlet, Y_olefin
- **Selectivities:** S_CH4, S_C2, S_C3, S_C4, S_C2–C4
- **Outlet Pressures:** P_CO2, P_H2, P_CO, P_H2O (out)

Generated entirely by our own Python ODE pipeline — not third-party data.

Dataset Analysis & Physical Observations

Checking that the synthetic data obeys known chemical-engineering trends



Both trends match expected catalysis behavior: higher T drives CO_2 conversion (endothermic RWGS) but shortens hydrocarbon chains.

Key Output Statistics (3,000 rows)

Variable	Min	Mean	Max
X_CO2 (%)	0.81	38.24	99.99
S_C2–C4 (%)	9.75	44.64	64.83
S_CH4 (%)	22.43	52.57	90.25
α	0.050	0.304	0.572
Y_olefin (%)	0.52	12.94	35.21

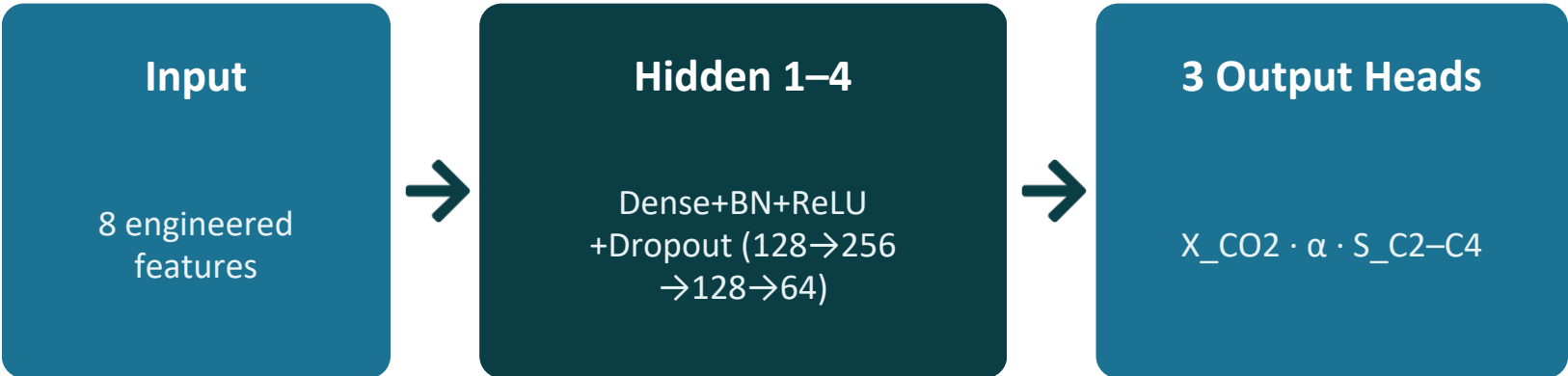


Peak olefin yield: 35.2%

Occurs at intermediate temperature (350–400°C) and high pressure (20–30 bar) — exactly the operating window predicted by ASF theory for the light-olefin regime ($\alpha \approx 0.40\text{--}0.65$).

Neural Network Surrogate Model

Shared backbone with separate output heads — 72,451 trainable parameters



Why separate output heads?

X_CO2 has a highly nonlinear temperature dependence, while α and S_C2–C4 are smoother — separate heads let the network allocate learning capacity per output.

Engineered Input Features

$T_K, P, H_2/CO_2, Keq, \log(Keq), \log(T_K), T_K^2, T_K \cdot P$

Training Configuration

Hyperparameter	Value
Optimizer	Adam, LR = 1×10^{-3}
LR schedule	ReduceLROnPlateau ($\times 0.5$)
Batch size	32
Max epochs	600 (early stop, patience 60)
Regularization	L2 = 10^{-4} + Dropout 5–10%
Data split	70% train / 15% val / 15% test

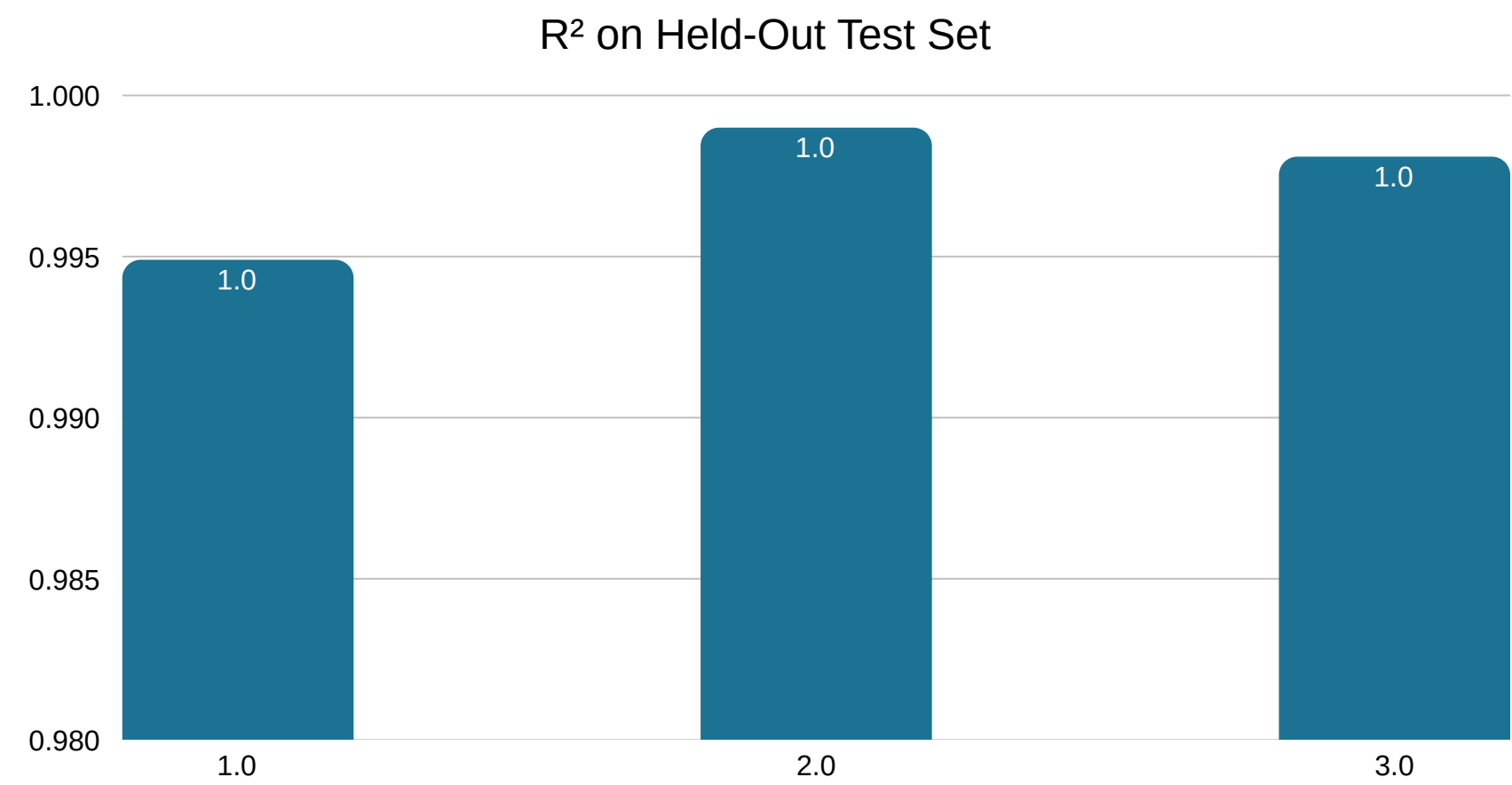
72,451

trainable parameters



Results & Model Performance

Evaluated on a held-out test set of 450 samples (15% of 3,000)



All three outputs exceed $R^2 = 0.99$ — the surrogate reproduces the ODE solver almost exactly.

Test-Set Error Metrics

Output	MAE	RMSE	Status
X_CO2 (%)	2.01%	1.98%	Excellent
α	0.0048	0.0051	Excellent
S_C2–C4 (%)	0.82%	0.85%	Excellent



Best operating point predicted by the surrogate

400°C, 25 bar → Y_olefin = 21.4%

High conversion and moderate α balance to maximize olefin yield. At 500°C, conversion is higher (88%) but α drops to 0.15, shifting selectivity toward CH₄ and lowering olefin yield (19.3%) — a physically correct, non-trivial trade-off the surrogate captures.

Anderson–Schulz–Flory (ASF) Model

Statistical chain-growth model governing Fischer-Tropsch product distribution

$$W_n = n(1 - \alpha)^2 \alpha^{n-1}$$

Reference

α range	Dominant product	Relevance
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0.65 – 0.80	Gasoline (C ₅ –C ₁₁)	–
> 0.80	Waxes (C ₁₂ +)	–

Dynamic α (Song et al. 2004, recalibrated)

We shifted the correlation's base term from 0.633 → 0.55 to target the light-olefin regime ($\alpha \approx 0.40\text{--}0.65$) instead of heavy hydrocarbons — consistent with Fe-K catalyst literature.

Two-Stage Product-Splitting Strategy

Why? Predicting C₂, C₃, C₄ directly from the NN can violate ASF consistency.

- 1

Stage 1 — NN

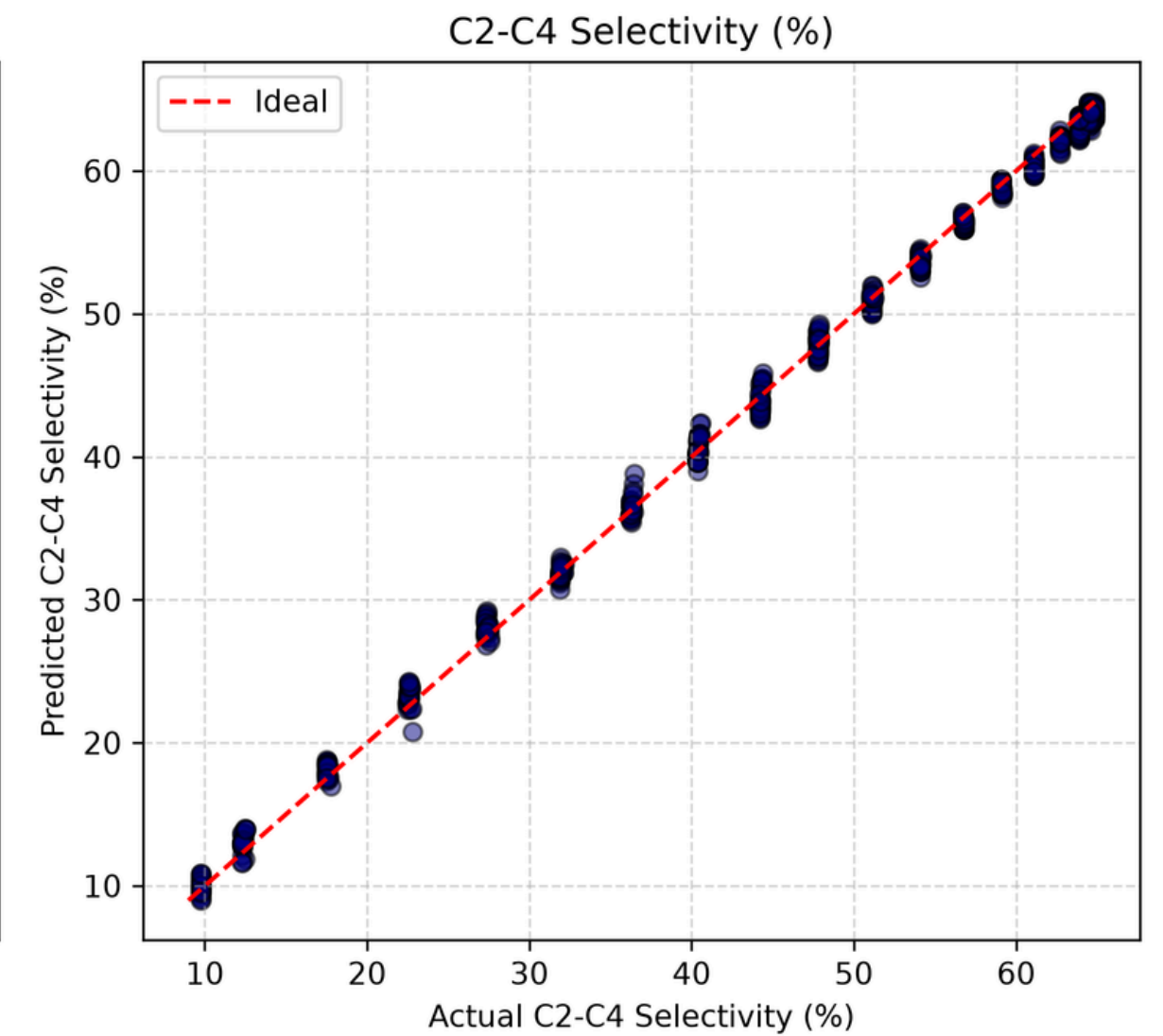
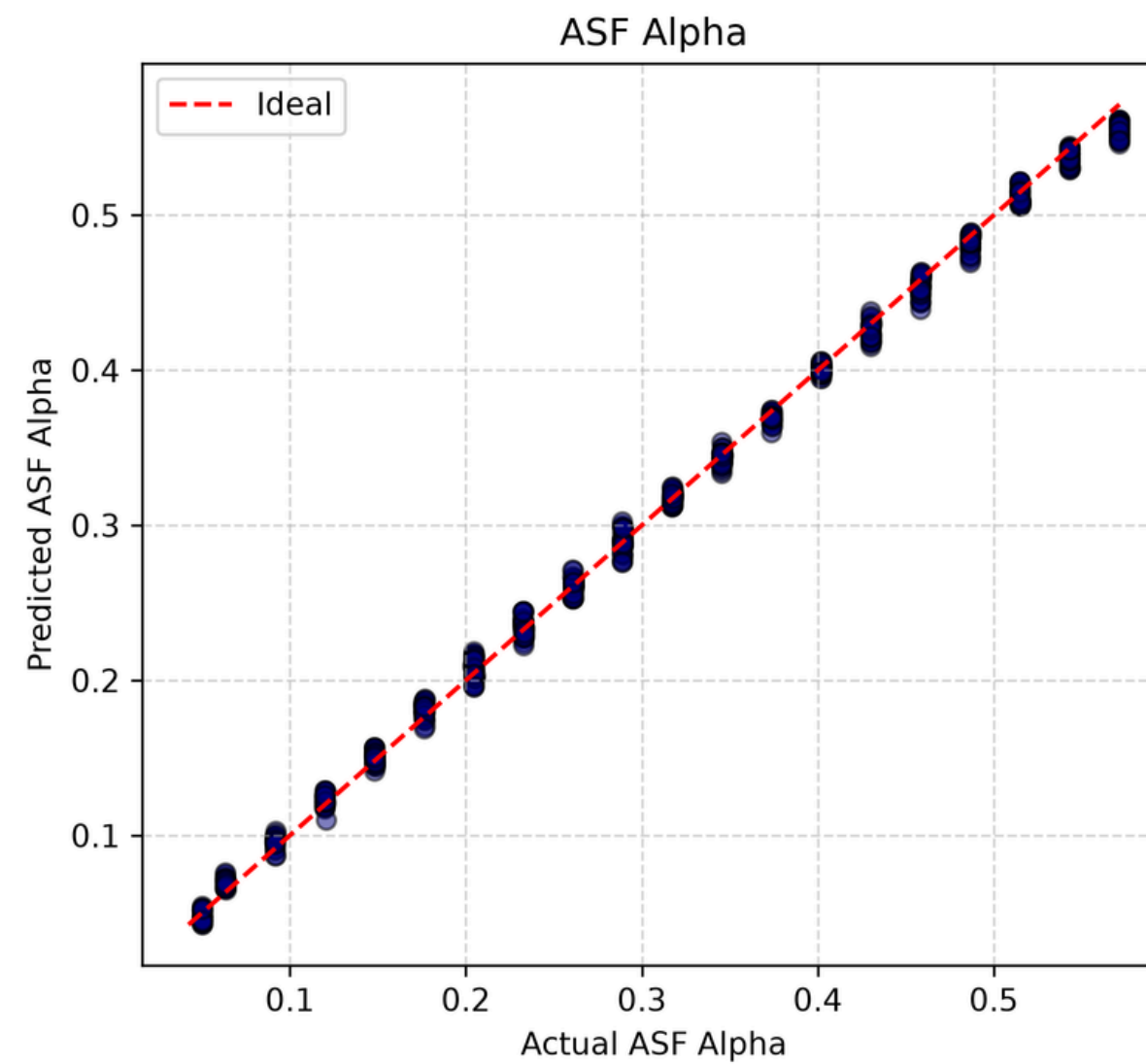
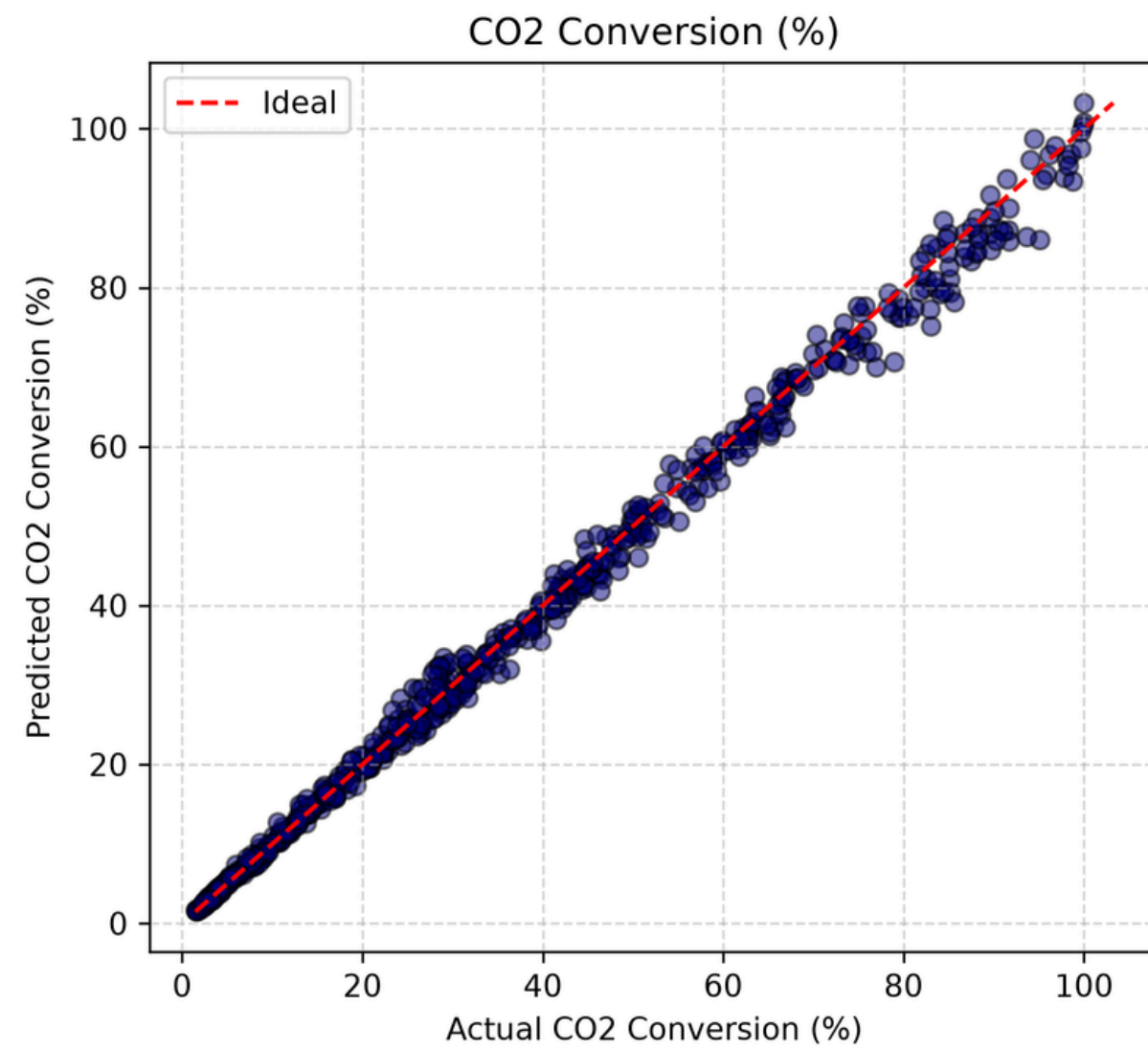
Predicts X_CO₂, α , and total S_C₂–C₄
- 2

Stage 2 — ASF

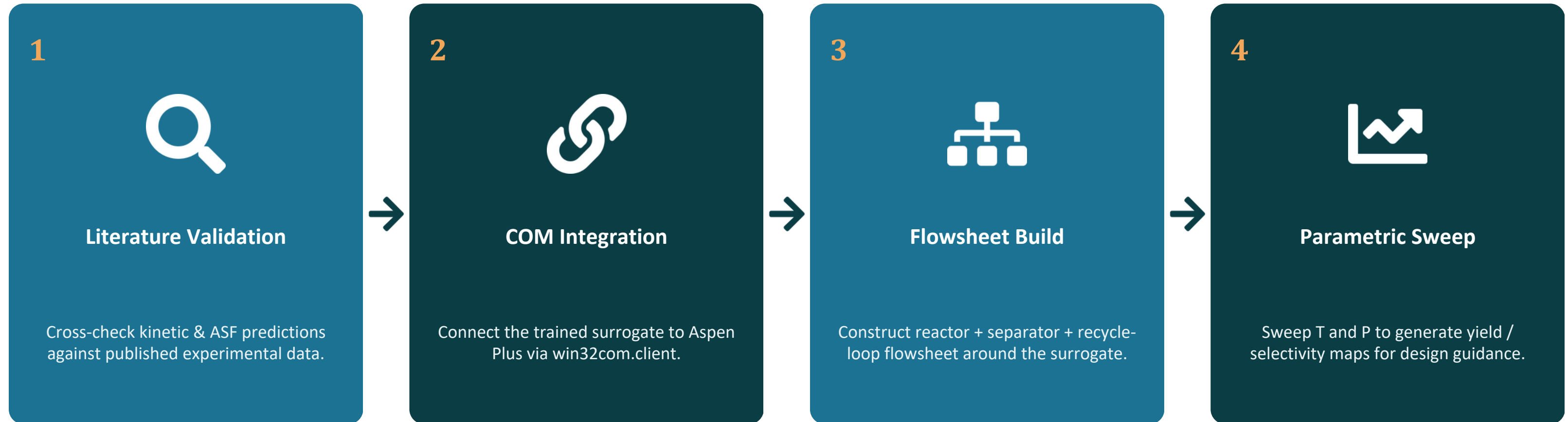
Splits S_C₂–C₄ into S_C₂, S_C₃, S_C₄ using Eq. (13)
- ✓

Guarantees the predicted product distribution always satisfies ASF physics — regardless of NN prediction error.

Graphs obtained

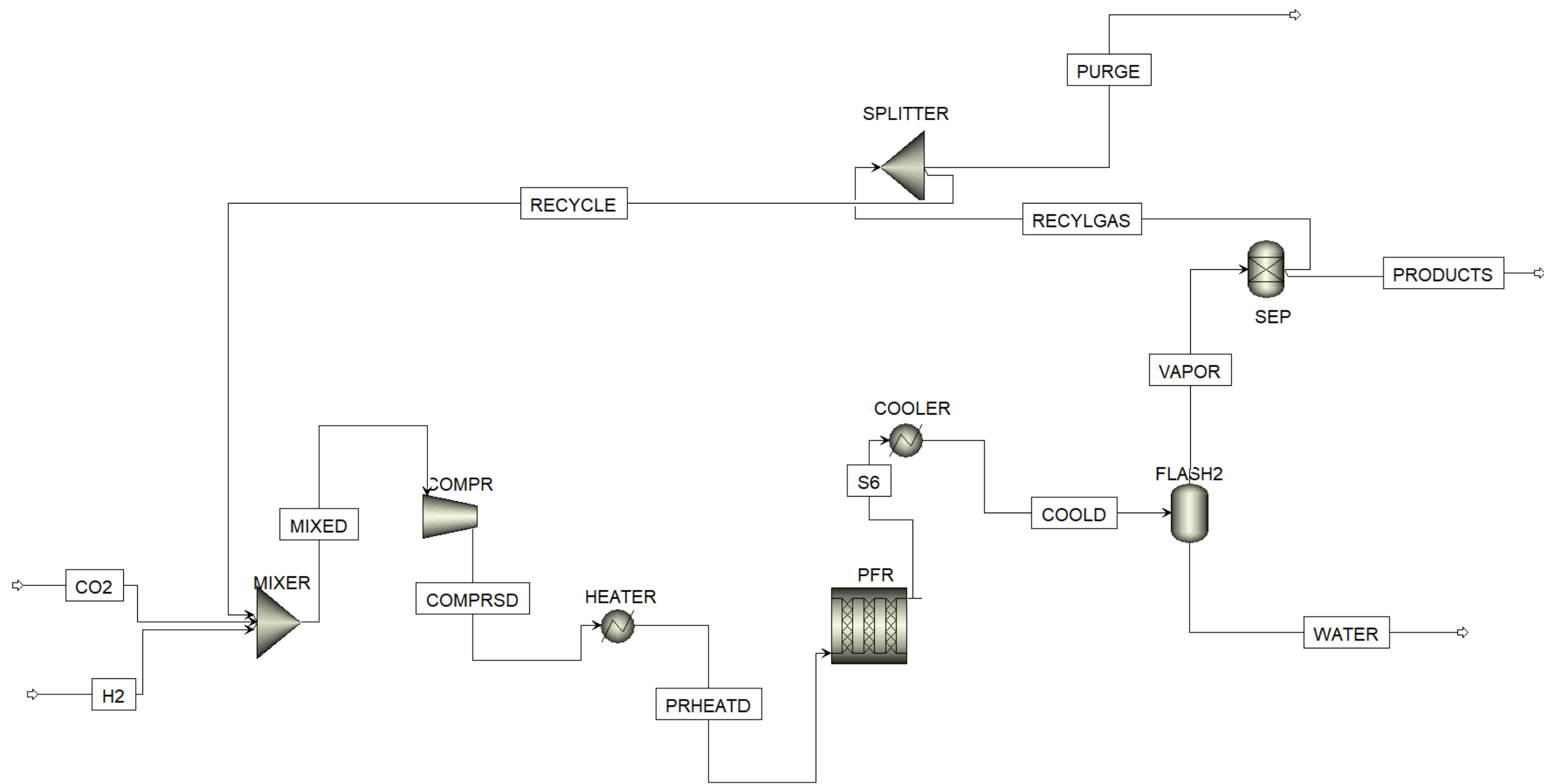


Aspen Plus & Full Digital Twin



End goal: a full plant-level digital twin where the surrogate provides millisecond reactor predictions inside an Aspen Plus flowsheet with recycle — enabling rapid yield/selectivity optimization that would be too slow with the ODE solver alone.

Aspen Plus Flowsheet



Aspen Plus COM Integration

win32client.com

```
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, i
compiler flags.

Launching Aspen Plus...
Aspen COM object created successfully.

Opening Aspen simulation...
Aspen file path:
C:\SAITEJA\Hybrid-Digital-Twin\SoC.bkp
Exists: True Open file in editor (ctrl + click)
Opening: C:\SAITEJA\Hybrid-Digital-Twin\SoC.bkp
Exists: True
Simulation loaded successfully.

Running Aspen simulation...
Simulation completed successfully.

Checking surrogate model...
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for
ill not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
Surrogate model loaded successfully.

Reading reactor feed stream (MIXED)...
Reactor feed successfully read.

Running surrogate reactor...
```

Python–Aspen Communication Workflow

- 1. **Launch Aspen Plus**
→ Create Aspen COM object and open the .bkp simulation
- 2. **Read Reactor Feed**
→ Extract T, P and H₂/CO₂ conditions from the Aspen feed stream
- 3. **Run Surrogate Model**
→ Python passes the reactor conditions to the trained neural-network surrogate
- 4. **Generate Reactor Outputs**
→ Predict CO₂ conversion, ASF α and C₂–C₄ selectivity
- 5. **Update Aspen Reactor Stream**
→ Write the predicted outlet composition/flow information back into Aspen
- 6. **Continue Flowsheet Simulation**
→ Aspen performs downstream cooling, separation, recycle and purge

Aspen Plus Stream Results

Material	Heat	Load	Work	Power	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	
		Units	CO2	COMPRSD	COOLD	H2	MIXED	PRHEATD	PRODUCTS	PURGE
▶	Average MW		44.0098	23.022	20.0955	2.01588	23.022	23.022	17.3555	27.8922
▶	➕ Mole Flows	kmol/hr	25	315.752	999.288	75	315.752	315.752	249.405	11.3561
▶	➕ Mole Fractions									
▶	➖ Mass Flows	kg/hr	1100.25	7269.25	20081.2	151.191	7269.25	7269.25	4328.55	316.747
▶	CO2	kg/hr	1100.25	1100.25	0	0	1100.25	1100.25	0	0
▶	HYDROGEN	kg/hr	0	151.191	0	151.191	151.191	151.191	0	0
▶	ETHYLENE	kg/hr	0	5.33078	280.602	0	5.33078	5.33078	274.978	0.28059
▶	PROPYLEN	kg/hr	0	0.882466	92.9001	0	0.882466	0.882466	91.9697	0.0464494
▶	METHANE	kg/hr	0	34.4253	3624.15	0	34.4253	34.4253	3587.8	1.81202
▶	CO	kg/hr	0	5976.77	6622.9	0	5976.77	5976.77	331.144	314.587
▶	WATER	kg/hr	0	0.30445	9449.58	0	0.30445	0.30445	31.7293	0.0160249
▶	1-BUT-01	kg/hr	0	0.104887	11.0416	0	0.104887	0.104887	10.9312	0.00552082
▶	➕ Mass Fractions									
▶	Volume Flow	l/min	10272.2	17051.4	10294.2	31001.4	133402	13729.5	5188.77	245.191
▶	➕ Vapor Phase									

- CO₂ and H₂ are fully consumed in the PRODUCTS stream
- Products contain C₂–C₄ olefins: ethylene, propylene and butylene
- Recycle and purge streams enable process continuity
- Aspen provides component-wise flow rates and overall stream properties for downstream process analysis

Challenges Faced

Technical obstacles encountered and how we resolved them



Aspen Plus

Integration of the surrogate reactor with the separation and recycle loop resulted in remaining mass-balance inconsistencies. Stream connections and calculations are being refined.



Recalibrating α Correlation

Song et al. (2004) targets heavy hydrocarbons ($\alpha \approx 0.7\text{--}0.9$). We had to shift the base term ($0.633 \rightarrow 0.55$) to land in the light-olefin window, validated against Fe-K literature.



Python–Aspen COM Integration

Mapping Aspen stream variables to Python required careful handling of COM object paths and stream properties. Automated communication and surrogate execution were successfully established.



Physical Consistency vs. ML Flexibility

Direct NN prediction of individual C2/C3/C4 selectivities could break ASF physics. Solved with the two-stage NN+ASF split strategy.